HCIA-AI V4.0 Certification Practice Exam • Chapter 4

Module 4: AI Development Framework (100 Questions & Answer Key)

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Question 1

TRUE OR FALSE • 16 Points

PyTorch adopts a dynamic computational graph (Define-by-Run), which constructs the graph on the fly as operators are executed, enabling intuitive debugging with standard Python tools.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In static computational graphs (Define-and-Run), the graph structure is dynamically rebuilt at every execution step, allowing arbitrary changes to layer shapes during runtime without recompilation.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

In PyTorch, calling optimizer.zero_grad() is necessary before loss.backward() because gradients accumulate by default across successive backward calls.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In PyTorch, the nn.CrossEntropyLoss criterion requires users to manually pass data through an explicit nn.Softmax layer before feeding it into the loss function.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

A Tensor in deep learning frameworks is strictly limited to 2-dimensional matrices and cannot represent data with 3 or more dimensions.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

Eager execution executes operations immediately and returns concrete tensor values without needing an explicit Session.run() call.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In PyTorch, custom neural network modules do not need to inherit from torch.nn.Module and can execute without defining a forward() method.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The torch.utils.data.DataLoader class is used only for saving trained model weight checkpoints to the hard drive.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Model quantization converts low-bit INT8 representations into high-precision FP64 to significantly increase model file size and memory footprint.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In AI model persistence, the Checkpoint (.ckpt) file format uses Protocol Buffers to store training parameter values by default (and can optionally save model structure and training state) for resuming interrupted training or fine-tuning.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: False • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

Model pruning increases model complexity and expands file size by randomly adding duplicate layers to the neural network.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In PyTorch, calling model.eval() turns on Dropout and updates running statistics in Batch Normalization layers.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

ONNX (Open Neural Network Exchange) provides an open ecosystem and universal intermediate representation format allowing models to transfer across different AI frameworks.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

MindSpore is Huawei's open-source all-scenario AI development framework designed to support unified APIs across cloud, edge, and device environments.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Automatic differentiation in deep learning frameworks automatically computes exact derivative values without requiring developers to manually write analytical gradient equations.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

A PyTorch Tensor residing in CPU memory can perform direct hardware matrix multiplications with a Tensor residing in GPU (CUDA) memory without explicit device transfer.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In PyTorch, setting tensor.requires_grad = False signals to the autograd engine that gradients must be actively tracked and computed for that tensor.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

TensorFlow 2.0 adopted Eager Execution as its default operating mode to significantly improve ease of development and debugging.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In deep learning frameworks, operator fusion combines multiple successive computational kernels into a single kernel to minimize memory access overhead and improve execution efficiency.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

TFRecord is a binary file storage format designed in TensorFlow for efficient high-throughput sequential data reading.

A. True
B. False

Page Answers: **Q16**: False • **Q17**: False • **Q18**: True • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

When constructing a custom Dataset in PyTorch, developers must implement both the __len__() and __getitem__() methods.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

The torch.no_grad() context manager forces PyTorch to track and calculate gradients for all tensor operations during evaluation.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In PyTorch, .pt and .pth are the default file formats for saving complete models (architecture and parameters) or parameter dictionaries, whereas converting a PyTorch model to a generic .bin format requires explicit format conversion.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Hardware acceleration in modern AI frameworks is achieved only through CPU multi-threading, while GPUs and NPUs cannot be used.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In Huawei's AI full-stack architecture, CANN (Compute Architecture for Neural Networks) provides high-performance operator libraries directly interfacing with Ascend AI chips.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI deep learning framework?

- **A.** MySQL
- **B.** TensorFlow
- **C.** PyTorch
- **D.** MindSpore

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a core characteristic of PyTorch?

- **A.** Mandatory static compilation of entire code into C++ sessions before executing single operators
- **B.** Native Pythonic development experience supporting standard control flow and interactive debuggers
- **C.** Seamless acceleration across different hardware targets through unified CUDA and NPU backends
- **D.** Dynamic computational graph construction where graphs are built on the fly

Question 28

SINGLE-ANSWER QUESTION • 16 Points

What is used in computational graphs to represent operations (such as matrix multiplication and convolution)?

- **A.** Edges
- **B.** Nodes
- **C.** Weights
- **D.** File descriptors

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In a computational graph, what do the directed edges connecting nodes represent?

- **A.** Operators
- **B.** Callbacks
- **C.** Tensors
- **D.** Optimizers

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is specifically responsible for automatic differentiation of tensors?

- **A.** torch.multiprocessing
- **B.** torch.distributed
- **C.** torch.autograd
- **D.** torch.optim.lr

Page Answers: **Q26**: A • **Q27**: A • **Q28**: B • **Q29**: C • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch calculates the backward gradients of a loss scalar with respect to all tracked leaf tensors?

- **A.** model.forward()
- **B.** loss.backward()
- **C.** tensor.zero_grad()
- **D.** optimizer.step()

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch updates the model parameters based on their computed gradients?

- **A.** loss.backward()
- **B.** model.eval()
- **C.** optimizer.step()
- **D.** torch.cuda.empty_cache()

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What happens if a developer forgets to call optimizer.zero_grad() at the beginning of each training iteration in PyTorch?

- **A.** The interpreter raises a fatal runtime exception and halts execution immediately on the forward pass
- **B.** Gradients from the current batch accumulate onto historical gradients, corrupting parameter updates
- **C.** The model skips forward propagation completely and freezes all parameter tensors at zero
- **D.** PyTorch automatically halves the optimizer learning rate after every ten training batches

Question 34

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, how do you move a model and input tensors to GPU for accelerated processing?

- **A.** model.compile()
- **B.** model.transfer()
- **C.** model.serialize()
- **D.** model.cuda()

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which PyTorch package contains popular computer vision datasets, model architectures, and image transformation utilities?

- **A.** torchaudio
- **B.** torchvision
- **C.** torchrec
- **D.** torchtext

Page Answers: **Q31**: B • **Q32**: C • **Q33**: B • **Q34**: D • **Q35**: B

Question 36

SINGLE-ANSWER QUESTION • 16 Points

In the PyTorch torchvision.transforms module, which transformation converts a PIL Image or NumPy array into a floating-point Tensor normalized to [0.0, 1.0]?

- **A.** transforms.Grayscale()
- **B.** transforms.Resize()
- **C.** transforms.Normalize()
- **D.** transforms.ToTensor()

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the collate_fn argument in PyTorch DataLoader?

- **A.** Merging a list of individual sampled dataset elements into a formatted, stacked mini-batch tensor
- **B.** Compiling high-performance CUDA operator kernels into native machine instructions ahead of time
- **C.** Serializing and exporting trained model architectures into cross-platform ONNX runtime binary formats
- **D.** Deleting unreferenced and redundant raw media files from external network-attached storage disks

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Define-by-Run' design pattern used in modern deep learning frameworks?

- **A.** The computational graph is dynamically constructed on the fly as each individual operator executes
- **B.** The model must be exported to static ONNX binary format before forward pass execution
- **C.** The execution graph is statically declared and frozen in advance before any data tensors can flow
- **D.** Operators are defined and orchestrated through external relational database schemas and SQL queries

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a major advantage of Static Computational Graphs?

- **A.** Direct integration with native Python if-else conditions and standard loop statements during runtime
- **B.** Total elimination of compilation phases, enabling instant execution of newly modified operator lines
- **C.** Native line-by-line interactive debugging support using standard Python debugging utilities like pdb
- **D.** Extensive ahead-of-time whole-graph optimizations, memory reuse planning, and easy embedded deployment

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which model compression method involves removing less important weight parameters (setting them to zero) based on magnitude or sensitivity?

- **A.** Model Quantization
- **B.** Knowledge Distillation
- **C.** Model Pruning
- **D.** Data Augmentation

Page Answers: **Q36**: D • **Q37**: A • **Q38**: A • **Q39**: D • **Q40**: C

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In Knowledge Distillation, what does the compact Student model learn from during training?

- **A.** Soft target probabilities from the larger Teacher model alongside hard ground-truth target labels
- **B.** Low-level operating system kernel diagnostic logs recorded during hardware boot-up sequences
- **C.** Zeroing out random activation channels during inference passes
- **D.** Raw Python source code comments and docstrings parsed directly from external code repositories

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What type of quantization converts 32-bit floating-point weights (FP32) to 8-bit integers (INT8)?

- **A.** Binarized Weight Pruning
- **B.** Unstructured Distillation
- **C.** INT8 Quantization
- **D.** FP64 Double Precision

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which PyTorch module implements a two-dimensional convolutional layer?

- **A.** torch.nn.Conv2d
- **B.** torch.nn.BatchNorm1d
- **C.** torch.nn.Linear
- **D.** torch.nn.Embedding

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is used to define a fully connected (dense) linear layer?

- **A.** torch.nn.Linear
- **B.** torch.nn.Dropout2d
- **C.** torch.nn.Conv1d
- **D.** torch.nn.RNNCell

Question 45

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for saving only the learned parameters (weights and biases) of a model?

- **A.** torch.save(model, PATH)
- **B.** json.dump(model)
- **C.** pickle.dump(model)
- **D.** torch.save(model.state_dict(), PATH)

Page Answers: **Q41**: A • **Q42**: C • **Q43**: A • **Q44**: A • **Q45**: D

Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for restoring learned parameters into an instantiated model architecture?

- **A.** model.restore_all_parameters
- **B.** model = torch.deserialize_weights
- **C.** model.load_state_dict
- **D.** torch.compile_and_mount_weights

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Why is model.eval() called prior to running validation or testing on a PyTorch neural network?

- **A.** It sets all gradient accumulators to zero and clears temporary intermediate autograd computation buffers
- **B.** It recompiles the Python model into optimized native C++ assembly binaries for rapid standalone execution
- **C.** It unmounts the training dataset partitions from physical storage to maximize available operating system RAM
- **D.** It switches Dropout and BatchNorm to evaluation mode, disabling random dropping and freezing statistics

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Huawei's MindSpore framework, what is the unified programming format that supports both dynamic debugging and static acceleration?

- **A.** C-only mode and Fortran mode
- **B.** TCP mode and UDP mode
- **C.** SQL mode and NoSQL mode
- **D.** PyNative mode and Graph mode

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an open-source framework developed by Google that introduced XLA (Accelerated Linear Algebra) and automatic differentiation for Python/NumPy programs?

- **A.** Theano
- **B.** PyTorch
- **C.** Caffe
- **D.** JAX

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, what tensor operation changes the tensor view/shape without altering the underlying data in memory?

- **A.** tensor.view() or tensor.reshape()
- **B.** tensor.backward_propagation()
- **C.** tensor.to_cuda_device()
- **D.** tensor.item_to_scalar()

Page Answers: **Q46**: C • **Q47**: D • **Q48**: D • **Q49**: D • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which function concatenates a sequence of tensors along a specified existing dimension?

- **A.** torch.chunk()
- **B.** torch.squeeze()
- **C.** torch.cat()
- **D.** torch.split()

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of the torch.squeeze() function?

- **A.** Normalizing tensor values between 0 and 1
- **B.** Compressing the tensor with ZIP compression
- **C.** Adding a new dimension of size 1 at a specified position
- **D.** Removing all dimensions of size 1 from the tensor

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of torch.unsqueeze(dim) in tensor manipulation?

- **A.** Inserting a new singleton dimension at the specified tensor index position
- **B.** Deleting all negative numerical values across the tensor buffer and clipping to zero
- **C.** Transferring tensor memory buffers from CPU host RAM to GPU device memory partitions
- **D.** Computing the inverse matrix transformation across two-dimensional square tensor slices

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which of the following represents the correct sequence in a typical PyTorch training step?

- **A.** output = model(input) -> optimizer.step() -> loss.backward() -> optimizer.zero_grad() -> log_batch_metrics()
- **B.** loss.backward() -> optimizer.step() -> optimizer.zero_grad() -> output = model(input) -> loss = criterion(output, target)
- **C.** optimizer.step() -> loss.backward() -> output = model(input) -> optimizer.zero_grad() -> loss = criterion(output, target)
- **D.** optimizer.zero_grad() -> output = model(input) -> loss = criterion(output, target) -> loss.backward() -> optimizer.step()

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is commonly used to adjust the learning rate during training according to predefined epoch schedules?

- **A.** torch.autograd.grad
- **B.** torchvision.models
- **C.** torch.optim.lr_scheduler
- **D.** torch.utils.data.Sampler

Page Answers: **Q51**: C • **Q52**: D • **Q53**: A • **Q54**: D • **Q55**: C

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which tool in PyTorch allows serializing dynamic models into static TorchScript graphs for production deployment without Python dependencies?

- **A.** torch.nn.parallel or torch.cuda.amp
- **B.** torch.jit.trace or torch.jit.script
- **C.** torch.save_weights or torch.load_bin
- **D.** torch.optim.sgd or torch.optim.adam

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What role does CANN play in Huawei's Ascend AI software stack?

- **A.** A cloud-based web browser designed for visualizing distributed multi-node training metrics and logs
- **B.** The computing architecture and chip enablement layer, exposing optimized operator APIs
- **C.** A mobile operating system kernel tailored specifically for consumer wearable smart devices and phones
- **D.** A relational database management system storing user authentication tokens and private access keys

Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is Huawei's lightweight AI engine designed for ultra-fast, low-latency model inference on mobile and IoT edge devices?

- **A.** MindSpore Lite
- **B.** Kunpeng OS
- **C.** ModelArts Cloud
- **D.** GaussDB

Question 59

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a common high-performance deployment engine developed by NVIDIA for deep learning inference on GPUs?

- **A.** PyTorch Eager
- **B.** Matplotlib
- **C.** TensorRT
- **D.** Pandas

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Model Compression and Pruning techniques are correct? (Multiple Choice)

- **A.** Structured pruning removes entire contiguous channels or layers to achieve direct hardware acceleration
- **B.** Pruning permanently doubles the precision of weight tensors to 64-bit double precision
- **C.** Model pruning reduces parameter count and inference memory footprint while striving to preserve accuracy
- **D.** Pruning can only be performed before training and cannot be applied to pre-trained weights

Page Answers: **Q56**: B • **Q57**: B • **Q58**: A • **Q59**: C • **Q60**: AC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In PyTorch data loading pipelines using DataLoader, which parameters directly optimize throughput and CPU-GPU transfer efficiency?

- **A.** num_workers
- **B.** pin_memory
- **C.** batch_size
- **D.** learning_rate

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques improve data transfer speeds and pipeline throughput in deep learning training?

- **A.** Using page-locked pinned host memory
- **B.** Asynchronous multi-threaded data pre-fetching and batch queuing
- **C.** Writing intermediate mini-batches to uncompressed text CSV files between training steps
- **D.** Storing training data in optimized binary container formats

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following loss functions are commonly implemented in modern deep learning frameworks (MindSpore, PyTorch) for continuous regression tasks?

- **A.** Categorical Cross-Entropy Loss
- **B.** Mean Squared Error Loss
- **C.** Binary Cross-Entropy with Logits
- **D.** Smooth L1 Loss

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In AI compiler software stacks (such as Huawei CANN and TVM), which benefits are achieved by Operator Fusion?

- **A.** Significant reduction in intermediate tensor memory read/write access overhead
- **B.** Converting non-linear neural networks into closed-form linear equation systems
- **C.** Minimization of hardware kernel launch overhead and execution scheduling latency
- **D.** Doubling the physical floating-point register count on the processor silicon core

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following services and capabilities are provided by Huawei Cloud ModelArts AI development platform?

- **A.** Bare-metal FPGA bitstream synthesis and hardware register-transfer level logic routing
- **B.** Automated data labeling, dataset management, and feature store integration
- **C.** Distributed large-scale model training with automated hyperparameter optimization
- **D.** One-click model deployment, real-time online inference API hosting, and batch serving

Page Answers: **Q61**: ABC • **Q62**: ABD • **Q63**: BD • **Q64**: AC • **Q65**: BCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary functions provided by an AI Development Framework?

- **A.** High-level front-end programming APIs for defining neural network layers
- **B.** Automatic differentiation and gradient computation engine
- **C.** Model optimization, serialization, and deployment tools
- **D.** Hardware abstraction and computation scheduling

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are defining characteristics of Eager Execution (Define-by-Run) mode in deep learning frameworks?

- **A.** Immediate execution of operations with direct numerical feedback
- **B.** Mandatory ahead-of-time compilation of the complete computational graph
- **C.** Seamless line-by-line debugging using standard Python debugging tools like pdb
- **D.** Immutability of network layer topologies during runtime execution

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following phases are included in the end-to-end deep learning engineering lifecycle?

- **A.** Model validation, evaluation, and hyperparameter tuning
- **B.** Data preparation, cleansing, and preprocessing pipelines
- **C.** Model architecture construction and training loop execution
- **D.** Manual hexadecimal byte rewriting of pre-trained binary weights before serving

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary attributes of a PyTorch Tensor?

- **A.** requires_grad
- **B.** shape
- **C.** dtype
- **D.** device

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware accelerators are primarily targeted by modern deep learning frameworks for high-throughput matrix computation?

- **A.** Legacy Optical Disc Drives
- **B.** Serial RS-232 Port Controllers
- **C.** Graphics Processing Units
- **D.** Neural Processing Units

Page Answers: **Q66**: ABCD • **Q67**: AC • **Q68**: ABC • **Q69**: ABCD • **Q70**: CD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Static Computational Graphs are true?

- **A.** It enables extensive compiler-level global graph optimizations and operator fusions
- **B.** It allows native interactive debugging using standard Python print statements inside the graph execution
- **C.** It is typically easier to export to standalone embedded C++ runtimes
- **D.** The graph topology is built once and executed repeatedly

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Dynamic Computational Graphs are true?

- **A.** The computational graph is built dynamically in memory during each forward pass
- **B.** Debugging is very straightforward because inspection is synchronous
- **C.** Dynamic control flow is fully supported
- **D.** It completely eliminates the need for GPU drivers

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

When implementing, training, and saving the LeNet CNN architecture in PyTorch (as presented in the HCIA-AI V4.0 training material and lab guide), which of the following statements are correct?

- **A.** Convolutional feature maps are flattened via x = x.view(-1, 16*5*5) before entering nn.Linear(16*5*5, 120)
- **B.** Pooling operations such as nn.AvgPool2d(kernel_size=2, stride=2) or torch.max_pool2d(x, 2) downsample feature maps without learnable weights
- **C.** Saving only learned weights with torch.save(net.state_dict(), path) requires instantiating net = LeNet() before calling net.load_state_dict(...)
- **D.** Calling net.eval() automatically overwrites saved .pth checkpoint files with random uninitialized weights

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common loss functions available in PyTorch (torch.nn)?

- **A.** nn.BCELoss
- **B.** nn.CrossEntropyLoss
- **C.** nn.MSELoss
- **D.** nn.L1Loss

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following optimization algorithms are implemented in PyTorch's torch.optim module?

- **A.** torch.optim.SGD
- **B.** torch.optim.Adam
- **C.** torch.optim.Softmax
- **D.** torch.optim.RMSprop

Page Answers: **Q71**: ACD • **Q72**: ABC • **Q73**: ABC • **Q74**: ABCD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

When creating a custom Dataset class in PyTorch, which magic methods MUST be implemented?

A. __len__() (returning the total size of the dataset)
B. __init_weights__() (initializing random network parameters)
C. __backward__() (computing analytic loss gradients)
D. __getitem__() (fetching an individual data sample by index)

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

According to the HCIA-AI V4.0 Model File specification, which of the following statements accurately describe standard model file formats?

A. Checkpoint (.ckpt) uses Protocol Buffers to store parameters by default and is used to resume interrupted training or fine-tune
B. ONNX (.onnx) provides a general model expression for model migration between frameworks or execution on TensorRT
C. Generic binary (.bin) is a common binary format requiring explicit conversion from PyTorch models
D. PyTorch (.pt or .pth) is PyTorch's default model file format for saving and loading complete models or weights

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which execution modes are natively supported by Huawei's MindSpore AI development framework?

A. Graph Mode
B. PyNative Mode
C. Assembly Mode
D. Flash Emulation Mode

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are valid operations for altering tensor shapes in PyTorch?

A. tensor.transpose()
B. tensor.squeeze()
C. tensor.reshape()
D. tensor.view()

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary benefits of wrapping inference execution inside the with torch.no_grad(): context block?

A. Automatically pruning 50% of the neural network weights
B. Disabling autograd tracking to greatly reduce GPU memory consumption
C. Accelerating forward pass execution by skipping computation graph generation
D. Eliminating the necessity of loading pretrained model weights into RAM

Page Answers: **Q76**: AD • **Q77**: ABCD • **Q78**: AB • **Q79**: ABCD • **Q80**: BC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following operations occur during model backpropagation and optimization?

- **A.** Storing gradients into the .grad attribute of parameter tensors
- **B.** Erasing the original dataset files from storage
- **C.** Updating weight parameters in the negative gradient direction using the learning rate
- **D.** Computing partial derivatives of the loss function via the chain rule

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are image transformations available in torchvision.transforms?

- **A.** transforms.RandomHorizontalFlip
- **B.** transforms.ColorJitter
- **C.** transforms.Normalize
- **D.** transforms.RandomCrop

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major advantages of Post-Training Quantization (e.g., FP32 to INT8)?

- **A.** Greatly reduced model memory and storage footprint on disk
- **B.** Guaranteed zero accuracy loss across all complex downstream tasks
- **C.** Doubling the quantity of trainable floating-point parameters
- **D.** Accelerated inference throughput and reduced latency on edge hardware

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common distributed training methods supported by modern AI frameworks?

- **A.** Standalone single-core CPU execution without distributed communication
- **B.** Data Parallelism
- **C.** Model or Tensor Parallelism
- **D.** Pipeline Parallelism

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's full-stack AI development ecosystem?

- **A.** Ascend AI processors
- **B.** ModelArts development platform
- **C.** MindSpore AI framework
- **D.** CANN compute architecture

Page Answers: **Q81**: ACD • **Q82**: ABCD • **Q83**: AD • **Q84**: BCD • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are reasons why AI development frameworks are necessary for deep learning?

- **A.** Hand-calculating and hardcoding analytic Jacobian matrices for every layer
- **B.** Frameworks automatically eliminate all need for domain training data collection
- **C.** Frameworks provide unified multi-hardware acceleration
- **D.** Frameworks provide automated memory management and memory reuse across execution graphs

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PyTorch DataLoader are correct?

- **A.** It supports multi-process data loading via num_workers
- **B.** It automatically designs neural network architectures
- **C.** It allows shuffling of samples before each epoch
- **D.** It supports automatic batching of data samples

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are widely adopted open standard formats/runtimes for cross-framework model deployment and inference?

- **A.** Open Neural Network Exchange
- **B.** HTML5 Web Document Format
- **C.** TensorRT or MindSpore Lite runtime formats
- **D.** Comma-Separated Values Tabular Format

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following layers in torch.nn require input dimension configurations during initialization?

- **A.** nn.Flatten
- **B.** nn.Conv2d
- **C.** nn.Linear
- **D.** nn.ReLU

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are advantages of Knowledge Distillation?

- **A.** Guarantees that the student model has zero weight parameters
- **B.** Does not require altering the hardware architecture of edge deployment devices
- **C.** Enables lightweight models to achieve accuracy close to large teacher models
- **D.** Significantly reduces inference latency on resource-constrained devices

Page Answers: **Q86**: CD • **Q87**: ACD • **Q88**: AC • **Q89**: BC • **Q90**: BCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In deep learning computational graphs, which of the following statements regarding nodes and edges are TRUE?

A. Nodes represent mathematical operations or mathematical functions
B. Edges represent the flow of multidimensional arrays between computational operations
C. Automatic differentiation engines traverse the graph in reverse topological order during backpropagation
D. Computational graphs can only execute sequential linear pipelines and prohibit branching

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Static and Dynamic Computational Graphs are correct?

A. Dynamic computational graphs construct the execution graph on-the-fly during runtime
B. Static computational graphs build the full symbolic graph structure prior to execution
C. Static computational graphs help compiler-level optimizations such as operator fusion and memory reuse
D. Modern AI frameworks can convert dynamic functions into optimized static graphs

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the core package responsible for automatic differentiation and gradient tracking is torch.____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, all custom neural network models and layers must inherit from the base class torch.nn.____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, the Checkpoint (____) file format uses Protocol Buffers to store network parameters for resuming interrupted training or fine-tuning.

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: autograd • **Q94**: Module • **Q95**: .ckpt (or ckpt)

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the forward computational execution of a neural network must be defined inside the def ____(self, x): method of nn.Cell.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, all neural network models and functional layers must inherit from the base class mindspore.nn.____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, learned network weights are saved to a checkpoint file on disk using the function mindspore.____(network, path).

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

To migrate a trained PyTorch (.pt/.pth) model between different AI frameworks or deploy it on an inference engine like TensorRT, developers convert it into the open ____ format.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the class from torch.utils.data used to combine a dataset with mini-batching, shuffling, and multi-threaded loading is ____.

Answer: _____

Page Answers: **Q96**: construct • **Q97**: Cell • **Q98**: save_checkpoint • **Q99**: ONNX (or .onnx) • **Q100**: DataLoader